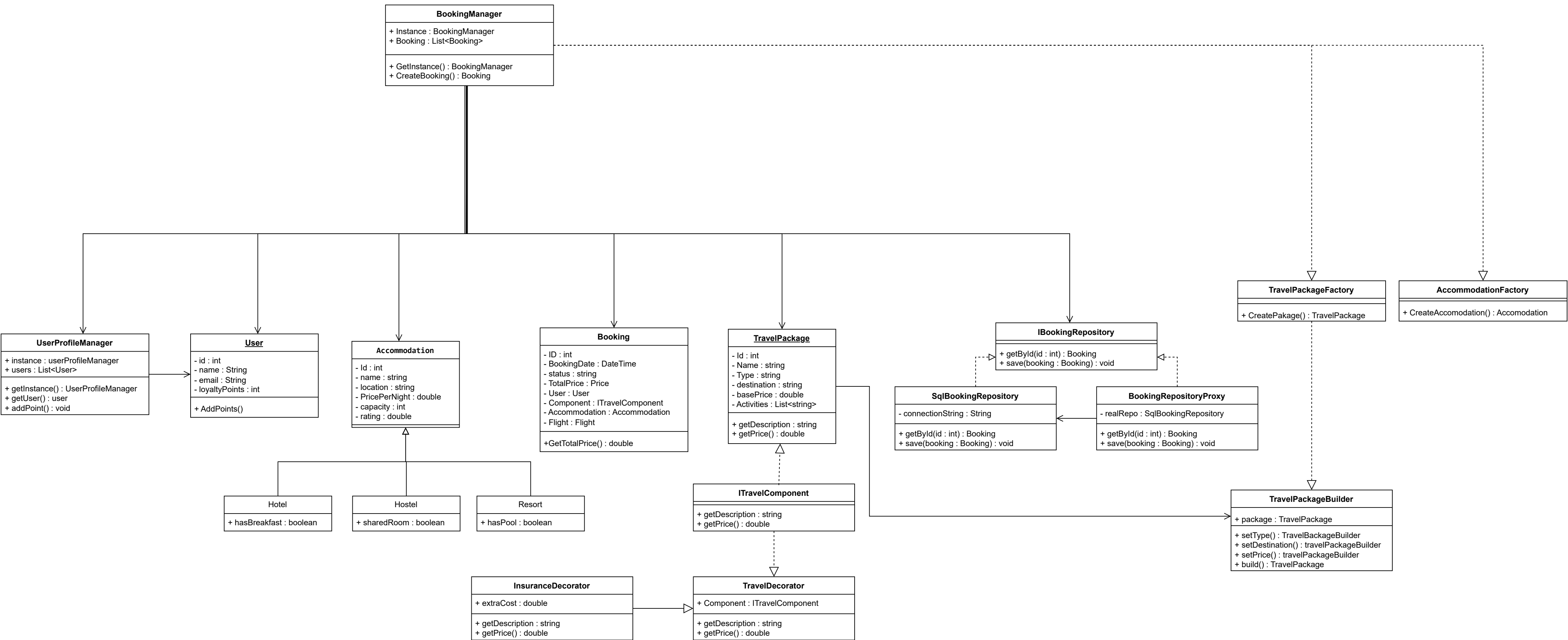




16. Travel Management System
Description: A desktop application to help travel agencies manage flight bookings, hotel reservations, and customer preferences.
• Singleton Pattern:
o Task: Booking Manager - A Singleton to ensure booking operations are handled consistently (flights, hotels, packages).
o Task: User Profile Manager - A Singleton to handle user details such as preferences, loyalty points, and booking history.
• Factory Pattern:
o Task: Factory for Travel Packages (e.g., Luxury, Adventure, Cultural) to create package objects based on user choice.
o Task: Factory for Accommodation Types (e.g., Hotel, Hostel, Resort) to create accommodation objects depending on the user's selection.
Patterns Used: Singleton, Factor

Singleton ==> BookingManager, UserProfileManager
Factory ==> TravelPackageFactory, AccommodationFactory
Builder ==> TravelPackageBuilder
Decorator ==> ITravelComponent + TravelPackage + TravelDecorator + InsuranceDecorator
Proxy ==> IBookingRepository + SqlBookingRepository + BookingRepositoryProxy

singleton
factory
builder
decorator
Proxy